

Research on Multi-object Tracking Algorithm Based on Person Recognition and Detection Fusion

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Abstract—To solve the problem of poor tracking performance in complex scenes such as scale change and frequent occlusion, a multi object tracking method based on the fusion of person detection and person recognition is proposed. Firstly, the improved DLA-34 network is used as the backbone network, and the residual block is improved to improve the object detection accuracy; Then a Re-ID module based on time graph convolution neural network is designed to effectively solve the problem of person identity switching and tracking object loss caused by occlusion and small object in video. Finally, the effectiveness of the algorithm is verified on MOT16 and MOT17 datasets, and the accuracy and tracking accuracy of multi- object tracking are improved. The experimental results show that the network has good detection and tracking effects.

Keywords—multi-object tracking; Graph convolution neural network; person re-identification

I. INTRODUCTION

The main task of multiple object tracking (MOT) is to detect, locate and track multiple interested objects simultaneously in a given video. In practical application scenarios, the multi-object tracking algorithm often faces many problems, such as mutual occlusion between persons caused by crowd congestion, frequent change of person posture, missed detection and false detection caused by persons with small objects in the distance, inaccurate object bounding box, and so on. Therefore, it is still a challenging task to propose an online multi-object tracking algorithm that is robust and has strong generalization ability in practical scenarios.

With the development of multi task learning, many researches have put forward excellent algorithms based on JDE integrated framework. Among them, the single step method of multi person tracking solves the problems of tracking accuracy and real-time through the framework of detector weighted recognition module embedding [1]. Video

based person recognition retrieves specific objects from video sequence images. When persons in video sequence are detected and tracked, occlusion between persons and similarity of object appearance will lead to fuzzy data association. To solve these problems, literature [2-3] alleviated the problem of data association ambiguity by weighted fusion of multiple clues, including motion information, appearance features and border box intersection and union ratio.

This paper proposes a tracking method based on the fusion of person detection and image convolution recognition based on depth learning. The main contributions are as follows: 1) Based on the cascade feature convolution module, improve the backbone network, improve the representation of features at different scales, and then improve the object detection accuracy. 2) A Re-ID module of Temporal Graph Convolution Neural (GCN) network is designed to effectively solve the problem of person identity switching and tracking object loss caused by occlusion and small object in video. 3) The validity and robustness of the algorithm are verified on MOT16 and MOT17 datasets, which provides a reasonable solution to the problem of occlusion and small object detection in the field of person multi- object tracking.

II. MATERIALS AND METHODS

In this paper, one shot method is used to add Re-ID module which is parallel to the detector head on the basis of CenterNet for anchor free object detection, so that it can share the feature extraction network with the detector. As shown in Figure 1, firstly, the improved deep layer aggregation network [4] (DLA-34) is used as the backbone network for feature extraction; Secondly, the object detection network is used to output the location information and embedded information of persons; At the same time, the person re recognition module based on time map convolution is adopted to improve the quality of feature vectors and enhance the Re-ID effect. Finally, simple Re-ID is used for data association.

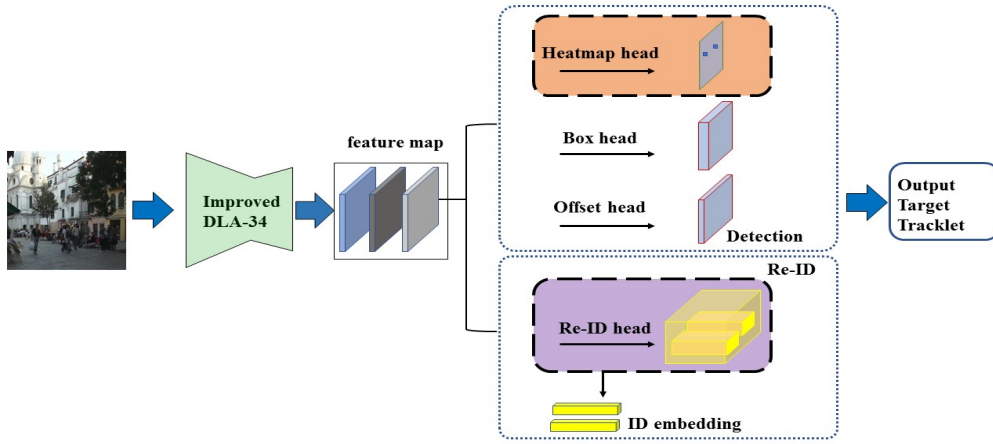


Fig.1. Model structure.

A. Backbone network

In order to meet the different requirements of feature extraction for detection and re recognition, this paper uses an improved DLA-34 network, as shown in Figure 2. This network contains many low dimensional features and high dimensional features, which can better integrate deep and shallow features, and extract target location and apparent information. First, the residual block of DLA-34 is integrated with the cascade features convolution (CFC) module. During the upsampling process, deformable convolution is used to replace the traditional convolution block, in order to obtain more accurate receptive fields in the process of generating feature maps of different scales.

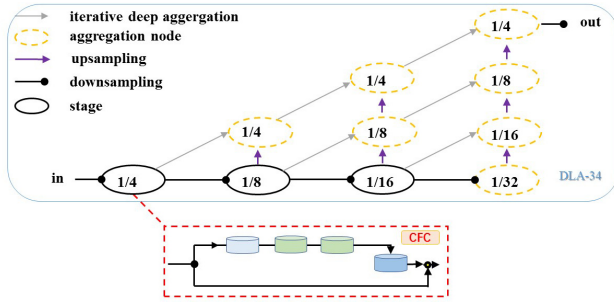


Fig.2. Schematic diagram of improved DLA-34 network.

The cascaded feature convolution module is composed of cascaded 1×3 and 3×1 convolution cores, as shown in Figure 3. Replace the 3×3 convolution core of the residual block Basic Block and Bottleneck in the original DLA-34 respectively, and use the BasicBlock integrated with the CFC module as the DLA-34 block to have better tracking effect. At the same time, according to the shape and size of the current target, adaptively adjust the position of the sampling points to extract the target features more accurately.

B. Detection branch

The feature map extracted by the backbone network is input to the detection branch, and the detection frame still

retains the branch of the original frame. The center coordinates, coordinate offsets, and length and width information of the tracking frame can be obtained from the three branches respectively. The information can be combined to obtain the detection result of the current frame for subsequent data association with the tracking track.

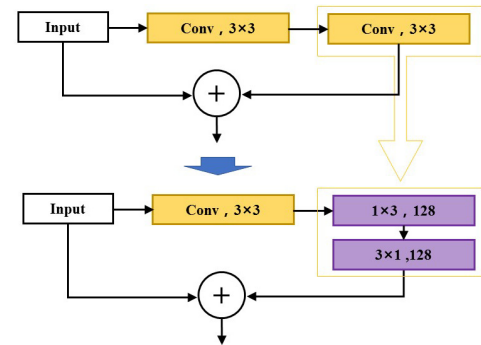
Focal Loss [5] function is used for central heat map loss L_{heat} , and the specific expression is as follows:

$$L_{heat} = -\frac{1}{N_{xy}} \sum_{xy} \begin{cases} (1 - \hat{M}_{xy})^\alpha \log(\hat{M}_{xy}), & M_{xy} = 1 \\ (1 - \hat{M}_{xy})^\beta \hat{M}_{xy}^\alpha \log(1 - \hat{M}_{xy}), & \text{other} \end{cases} \quad (1)$$

where, $\hat{M}_{xy}$ represents the value of the central heat map generated by the detection frame tag at point (x, y) , represents the value of the network predicted heat map at point (x, y) , and α, β are the preset parameters of Focal Loss.

The size and offset of the corresponding position evaluation are respectively expressed as $\hat{d}_i, \hat{b}_i$. Then implement L1 loss on two branches:

$$L_{box} = \sum_{i=1}^I \|b_i - \hat{b}_i\|_1 + \|d_i - \hat{d}_i\|_1. \quad (2)$$



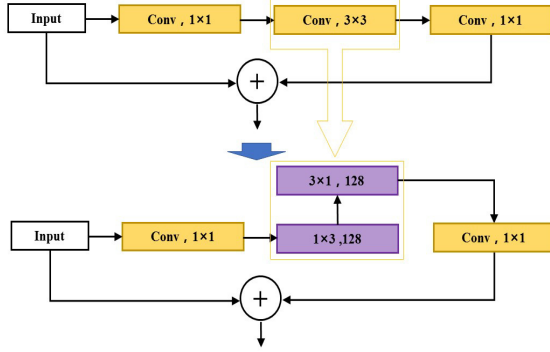


Fig.3. Replace the 3×3 convolution core of the residual block Basic Block and Bottleneck in the original DLA-34.

C. Figure Convolutional neural network person recognition branch

Because occlusion and noise impact will reduce the quality of ID embedding feature vectors, we use Graph evolution neural network (GCN) to model the time relationship between patches in different frames as patch feature supplement information while obtaining appearance features, so as to enhance the effect of re recognition.

We horizontally divide the feature map extracted from the backbone network into P patches, then pool each patch evenly, and then use time average pooling to aggregate the features of different frames to get f_s . At the same time, in order to explore and utilize the relationship between patches, x_i is defined as the feature of each patch, and the time relationship between patches is modeled. We use all patches to build a time graph $G(v, \varepsilon)$, and each video has N patches. The number of nodes in the graph is equal to the total number of video patches N , and the node $v_i = \{x_1, x_2, \dots, x_N\} \in v$. The relationship $e_{ij} = (v_i, v_j) \in \varepsilon$ between patches corresponds to the adjacent matrix $K \in \mathbb{Z}^{N \times N}$, and K is a key component of GCN, where each element reflects the pairing relationship of patches. Inspired by [6], $e(x_i, x_j)$ is used to represent the pairwise relationship between every two patches. Each element in K should be non-negative, and the coefficient is between 0-1. Therefore each line of K is normalized by

$$K_{(i,j)} = \frac{e^2(x_i, x_j)}{\sum_{j=1}^N e^2(x_i, x_j)}. \quad (3)$$

Finally, we can get that the adjacent matrix corresponding to graph $G(v, \varepsilon)$ is

$$\hat{K} = \left\{ \sum_j [K_{(i,j)} + I_n] \right\}^{-1} [K_{(i,j)} + I_n], \quad (4)$$

where $I_n \in \mathbb{Z}^{N \times N}$ is the identity matrix. The number of GCN layers constructed is M , and the m -th layer ($1 \leq m \leq M$) is

$$X^m = \hat{K} X^{m-1} W^m, \quad (5)$$

where $X^{(m)} \in \mathbb{Z}^{N \times l_m}$ is the hidden feature of all patches in the m -th layer, and l_m is the feature dimension; $m=0$ is the initial patch feature; $W^{(m)} \in \mathbb{Z}^{l_{m-1} \times l_m}$ is the parameter matrix to be learned. After each layer of graph convolution, add a Layer Normalization [7] layer and a Leaky ReLU (negative input slope $\alpha = 0.1$). Finally, we perform the maximum pooling operation on X^M . Therefore, for each video, we can obtain its time GCN characteristics: $f_t \in \mathbb{Z}^{1 \times l_m}$. Re-ID outputs a feature map with the size of $128 \times 272 \times 152$. Each feature point contains 128 feature vectors.

We use the batch hard triplet loss function [8] and softmax cross-entropy loss function to train the network. The two loss functions are recorded as $L_{triplet}$ and $L_{softmax}$ respectively. We calculate the $L_{triplet}$ and the $L_{softmax}$ of f_s and f_t respectively. In our experiment, we connect f_t and f_s in series as the final feature. Finally, we use the final feature to calculate the softmax cross-entropy loss. Therefore, the total loss L_{id} is a combination of these two losses:

$$L_{id} = L_{triplet}^t + L_{triplet}^s + L_{softmax} \quad (6)$$

D. Multi task training

In order to train the detection task and Re-ID task at the same time, the uncertainty loss [9] dynamic balance task is used, and the calculation is as follows:

$$L_{det} = L_{heat} + L_{box}, \quad (7)$$

$$L = \frac{1}{2} \left(\frac{1}{\exp(\omega_1)} L_{det} + \frac{1}{\exp(\omega_2)} L_{id} + \omega_1 + \omega_2 \right), \quad (8)$$

among them, ω_1 and ω_2 are learnable parameters for balancing tasks, with initial values of -1.85 and -1.05 respectively.

E. Online association

This paper uses the standard online tracking algorithm for correlation [4]. First, the object detection frame and Re-ID feature vector of the input image are extracted through the network, and then the cost matrix is calculated according to the cosine distance of Re-ID feature. At the same time, motion information is fused, Kalman Filter [10] and Mahalanobis Distance are used to exclude matches that are far away, and Hungarian algorithm [11] is used to complete the first match. For the unmatched track, the IoU cost matrix is calculated according to its detection frame with the unmatched object, and then the Hungarian algorithm is used to complete the second matching. Finally, update the track information, create a new track for the unmatched object, and record the unmatched track. When the number of times the existing track is not connected to the new object exceeds a certain threshold, the track is deemed to be over and will not be updated.

III. RESULTS & DISCUSSION

The experiment in this paper is based on the environment of Ubuntu 20.04 and Cuda10.2 and the Python deep learning framework. The hardware configuration includes three GPUs:

GTX 3070Ti (8GB video memory), and the programming language is Python 3.8.

A. Data sets and evaluation indicators

The dataset MOT17 is marked with identity information and object bounding box, which can train object detection and Re-ID branch networks at the same time. After training, the algorithm performance is evaluated in MOT16[12] and MOT17 test sets.

This experiment uses the evaluation algorithm of MOT Challenge Benchmark [12] to evaluate, and the evaluation indicators used are shown in Table 1. "↑" means the higher the index, the better.

B. Comparison with advanced MOT algorithm

To fully verify the performance of the algorithm in this paper, the MOT16 and MOT17 datasets are compared with the corresponding algorithms, as shown in Table 2-3. In MOT16, the algorithm in this paper has higher (MOTA) and (IDF1) than other algorithms. It has a higher track hit rate MT and a lower track loss rate ML. In MOT16, our method is superior to other existing methods in MOTA and IDF1. In MOT17, the more important MOTA index is 2.6% higher than the second place Fair MOT, and the IDF1 index is 4.9% higher than FairMOT. It can be seen that the proposed algorithm has achieved the first place in most indicators even in such a high complexity scenario. The results show that the algorithm has better tracking effect, while taking into account the real-time and accuracy. Both MOT16 and MOT17 data sets include multi-object tracking tasks in crowded scenes, which is consistent with the actual scene. Compared with MOT16, MOT17 has more accurate ground reality. The output results on the MOT17 test set in Figure 4 show that the proposed algorithm performs well in the crowded scene with many persons, which can not only maintain the correct boundary box, but also maintain the identity.

Table 1. Commonly used multi- object tracking evaluation indicators.

Evaluation Indicator	Description
MOTA	Multiple Object Tracking Accuracy (%) ↑
IDF1	Identification F1 Score (%) ↑
MT	Mostly Tracked Targets (%) ↑
ML	Mostly Lost Targets (%) ↓
IDs	Identity Switches ↓

Table 2: Comparison of different algorithms on MOT16 test set

Algorithm	MOTA↑	IDF1↑	MT↑	ML↓	IDS↓
JDE[13]	64.4	55.8	35.4%	20.0%	1544
CTracker[14]	67.6	57.2	32.9%	23.1%	1897
Tube_TK[15]	64.0	59.4	33.5%	19.4%	1117
Ours	77.3	77.5	52.2%	15.3%	632

Table 3: Comparison of different algorithms on MOT17 test set

Algorithm	MOTA↑	IDF1↑	MT↑	ML↓	IDS↓
Center Track[16]	67.8	64.7	34.6%	22%	3039
CTracker	66.6	57.4	32.2%	24.2%	5529
FairMOT[4]	73.7	72.3	43.2%	17.3%	3303
Ours	76.3	77.2	46.5%	17.1%	2241



Fig. 4. Output results of the proposed method on MOT17 test set.

The analysis and comparison on MOT16 and MOT17 benchmark datasets fully prove the effectiveness of the proposed method. Through analysis, it is concluded that the algorithm can better deal with the difficulties of small object detection in video, the difficulties of tracking objects caused by frequent occlusion during tracking, and the problems of person identity switching and tracking loss, which more intuitively shows the research value and practical significance of the algorithm in this paper.

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